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pyequa: Generation of quizzes based on relations declaration and data

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Software

- Review
- Repository
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Summary

pyequa is a Python package that provides a simple and efficient way to create multiple Cloze quizzes for Moodle (Moodle, 2025), allowing educators to generate dozens of interactive exercises quickly and easily.

An author needs this files:

- `exercise_model.md`: a markdown file (with text, figures, or tables) containing all variables declared like `{variable}`; later, in the generated quizzes, a variable can be "given" in text, or can be converted to a "fill-in-the-blank" (multiple choice or numerical);
- `data.csv`: a csv, or xlsx, file containing values or texts for variables; each row contains values that turns text in exercise model in a complete instance of some scientific scenario;
- `produce.py`: a Python file containing two string declarations :
  - `scenario_relations` : a multi-line string, with all relations between variables in a scenario; an example is 'Eq(mean, (x1+x2+x3)/3';
  - `variable_attributes`: type (numerical or multiple choice), tolerance (if numerical), optional output level (above 'mean' has level 2).

As an example follows, for basic statistics. The file `exercise_model.md` contains:

```
Consider this sample,
\(x\_1\)={x1}
\(x\_2\)={x2}
\(x\_3\)={x3}
and based on it, the following statistics are calculated:
\(\bar x\) = {mean}
\(s\_c^2\) = {variance}
variation coefficient = {cv}
```

Notice the "declarative" way of writting an exercise model.

Then, a data file, named `data.csv`, is created with columns 'x1', 'x2', 'x3', 'mean', 'variance', 'cv'. Values can be produced by hand or using code.

File `produce.py` contains this declarations:

```
scenario_relations = {
    "Eq(mean,  (x1+x2+x3)/3 )",
    "Eq(variance, ( (x1-mean)**2 + (x2-mean)**2 + (x3-mean)**2 )/2 )",
    "Eq(cv,   sqrt(variance)/mean)",
}

variable_attributes = {
```

```
    "x1":   {"type": "numerical", "tol": 0.05,  "givenvarlevel": 1},
    "x2":   {"type": "numerical", "tol": 0.05,  "givenvarlevel": 1},
    "x3":   {"type": "numerical", "tol": 0.05,  "givenvarlevel": 1},
    "mean": {"type": "numerical", "tol": 0.05,  "givenvarlevel": 2},
    "variance": {"type": "numerical", "tol": 0.05,  "givenvarlevel": 2},
    "cv":   {"type": "numerical", "tol": 0.05,  "givenvarlevel": 3},
}
```

A teacher can create several quizzes, in three main ways:

- evaluation quizzes,
- exploratory quizzes showing all combinations of "given" variables" versus "fill-in-the-blanks" variables, and
- "hard first" quizzes.

Evaluation quizzes . A teacher can produce questions for a moment of evaluation using Moodle "random question" (randomly selecting a quiz from a set of several similar quizzes). In `pyequa`, this is done inside `produce.py` with function `randomquestion_sameblanks()` as in the example:

```
randomquestion_sameblanks( fill_in_blanks_vars = {'mean', 'variance'},
                           number_of_problems_per_givenvars=4 )
```

The instruction produces 4 quizzes, all with two numerical fill-in-the-blanks (`mean` and `variance`) and values are given to all other variables (`x1`, `x2`, `x3`, `cv`).

Exploratory quizzes. A second method to produce quizzes helps the teacher to check diverse quizzes. All combinations of "given" variables" versus "fill-in-the-blanks" variables, in student text, are made. Eventually, a teacher can select some for class. In `pyequa`, this is done inside `produce.py` file with function `exploratory()` as in the example:

```
exploratory()  #no arguments
```

Hard first quizzes. An illustration of the concept "hard first" goes as follow: all calculators can easily compute the mean of a three element sample but if, in another quiz, the `mean` is given and only the element `x2` has to be calculated, the student must know how to use the arithmetic mean expression to get `x2`. The inverse problem are usually harder and it could be posed as a challenge to students. The instruction to do that is:

```
hard_first( max_number_of_problems=10,
            max_combinations_givenvars_per_easynesslevel=3,
            number_of_problems_per_givenvars=1)
```

It generates at most 10 problems, all with 1, 2 or 3 variables given (the others are fill-in-the-blanks), and for each combination of "given" variables" versus "fill-in-the-blanks" variables, only 1 quiz is generated.

Some notes:

- The output of this instructions is a folder `_output_` in two formats: markdown and xml for Moodle import.
- The authors use the package `pyequa` in VSCode and RStudio to produce quizzes for introductory courses in numerics and statistics.
- Data files (`data.csv` as explained previously) have been written by hand (carefully chosen interesting values or text strings for quizzes), or was coded in Python, R, or using Excel functions.
- If a column in data file has text, it will be implemented as a multiple choice field in Moodle like shown in the image:

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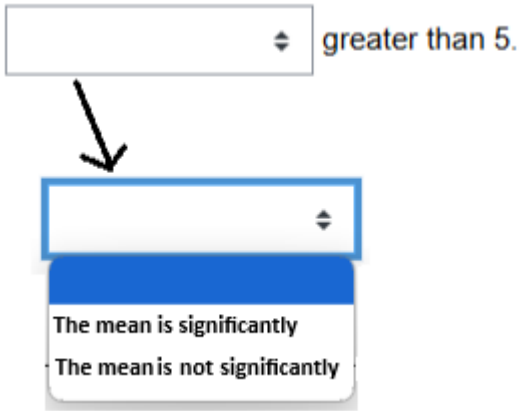

Figure 1: A fill-in-the-blanks variable with textual values is transformed into a multiple choice field.

Citations

Inspiration for this work came from a talk with authors of Araújo et al. (2016) where they dissect the notion of "Exercise composition". The same authors, in the related work Araújo et al. (2015), explore the case where a scenario is built of polynomials to be worked by Gröbner bases.

A previous work from first author (P. Cruz et al., 2012) was also an inspiration for this Python package.

Package is available at J. P. Cruz & Gomes (2025) that produces Moodle Cloze quizzes (Moodle, 2025). Package folder "examples" got several cases.

Statement of Need

Manual creation of parameterized quizzes, even with help of small chunks of code, can be a time-consuming and laborious process for teachers. A tool is introduced that eases the construction of quizzes for Moodle as these are an usual way for a student to practice concepts in STEM fields or being evaluated.

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Figure 2: Governo de Angola

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References

Araújo, I., Almeida, J. J., & Smirnov, G. (2016). Exercise composition: From environment properties to composed problems. In Á. Rocha, A. M. Correia, H. Adeli, L. P. Reis, & M. Mendonça Teixeira (Eds.), *New advances in information systems and technologies* (pp. 235–244). Springer International Publishing. https://doi.org/10.1007/978-3-319-31307-8_24

Araújo, I., Smirnov, G., & Almeida, J. J. (2015). Gröbner bases and mathematical exercises generation with nondetermined structure. *10th Iberian Conference on Information Systems and Technologies (CISTI) (2015)*, 1–5. https://doi.org/10.1109/CISTI.2015.7170359

Cruz, J. P., & Gomes, J. (2025). *Pyequa* (Version 0.0.1). https://github.com/jpcaveiro/pyequa

Cruz, P., Oliveira, P., & Seabra, D. (2012). Exercise templates with sage. *Journal of Statistical Software*, *5*(2), 37–44. https://doi.org/10.32513/tbilisi/1528768901

Moodle. (2025). *Embedded answers (cloze) question type*. https://docs.moodle.org/500/en/Embedded_Answers_(Cloze)_question_type.

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